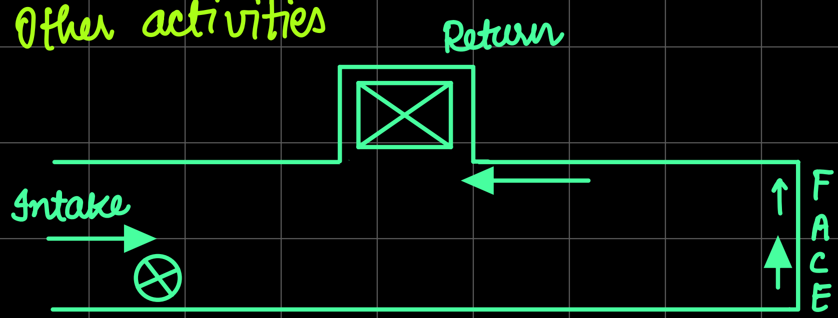
Drill jumbo operator helper

Manpower Distribution plan —

Sl. No.	Activity	Shift A (7AM-3PM)	Shift B (3PM - 11PM)	Shift C (11pm - 7AM)	Remarks
1.	Survey & Marking	1 sup + 2h + 1 MM + 1 FM			
2.	Drilling		1 $\boxed{\text{DJO}}$ + 1 $\boxed{\text{h}}$ + 1 MM + 1 FM		
3.	Charging & Blasting		1 B + 2h + 1 MM		(8:00 AM - 4:00 PM)
4.	Fume Clearence	No requirement of Manpower			
5.	Water spraying & loose dressing.			1 TM + 1 ATM + 1 MM + 1 FM	
6.	Loading & Hauling of blocked material			1 TM + 1 ATM + 1 MM + 1 FM 1 LHDO + 1 LPDTO + 1h	
7	Face preparation				
8.	Support		1 JhO + 2h + 1MM + 1 FM + 1 B + 2h	1 TM + 2h + 1 mm + 1 FM	
	Development Survey				
9.(i)	Fixing pegs and extending C/L & G/L	1 sur + 2 As + 1 MM + 1 FM			After 20m progress or advancement.
(ii)	Measuring opening size				

10.	Drainage	1RBO + 1h			
11.	Extension of Auxiliary services				
(i)	water pipe line & compressed air pipe line extension	1PF + 1h			
(ii)	Electrical cable extension & GEB extension	1ELEC + 1h + 1MM			
(iii)	Manhole	Jackhammer crew (1JHO + 2h) + 1MM + 1FM			
(iv)	Loading Bay	1PF + 2h			At an interval of 100m.
(v)	Ventilation duct extension				

	Activity	Shift Required
1.	Drilling + Charging & Blasting + WS + LD + Loading + Hauling + Face preparation	3 shifts for 1 round of blasting
2.	Other activities 	

Other activities	No. of shifts required
→ Surveying and Marking	01
→ Preparatory work : Compressed air pipeline, Water pipeline, Installation of Auxillary fan + Ventilation duct, Electrical cable laying + GEB	03
→ Loading Bay	11
→ Manhole + Support holes	1 shift / 20 m, 5 shifts for 100m
→ Peg setup + Measuring opening size	1 shift / 20 m, 5 shifts for 100m

MAIN ACTIVITIES :-

→ Drilling + Charging + Blasting + Fume clearance + WSLD + L&H	03 shifts per round of blasting
→ Back and side support (if required)	02 shifts after 3 rounds of blasting

Total shift needed for 100m

$$= \frac{100}{3.2 \times 0.9} \times 3 + \frac{100}{3.2 \times 0.9} \times \frac{2}{3} + 1 + 3 + 11 + 5 + 5$$

$$= 104 + 23 + 25$$

$$= 152$$

Total shift after considering B/d time + Unforeseen situation

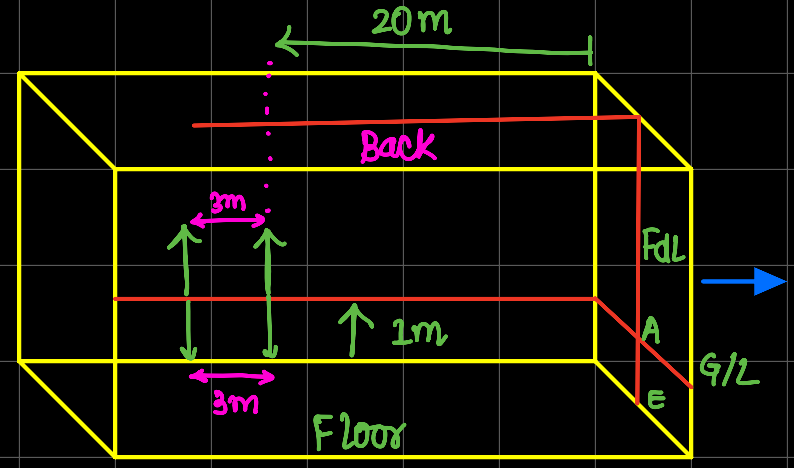
$$= 152 + 152 \times \frac{5}{100}$$

$$= 160$$

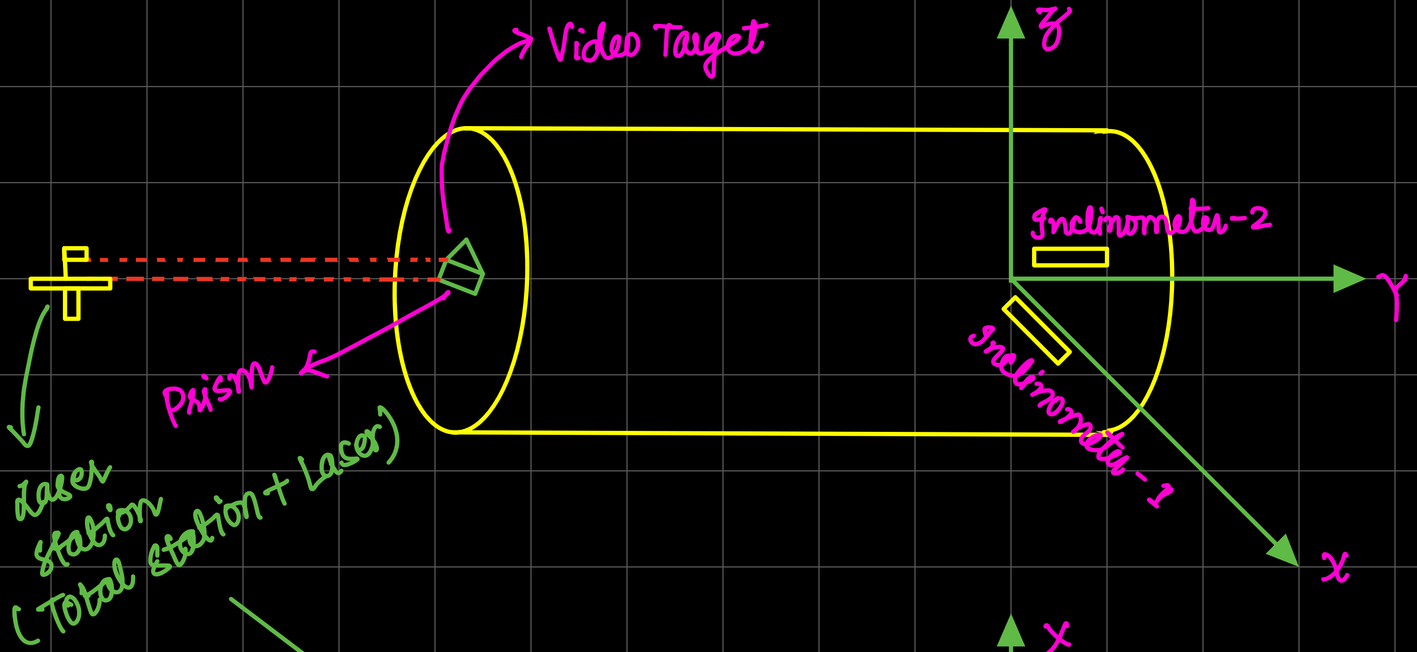
☆ Maintenance of direction and Gradient of a drift

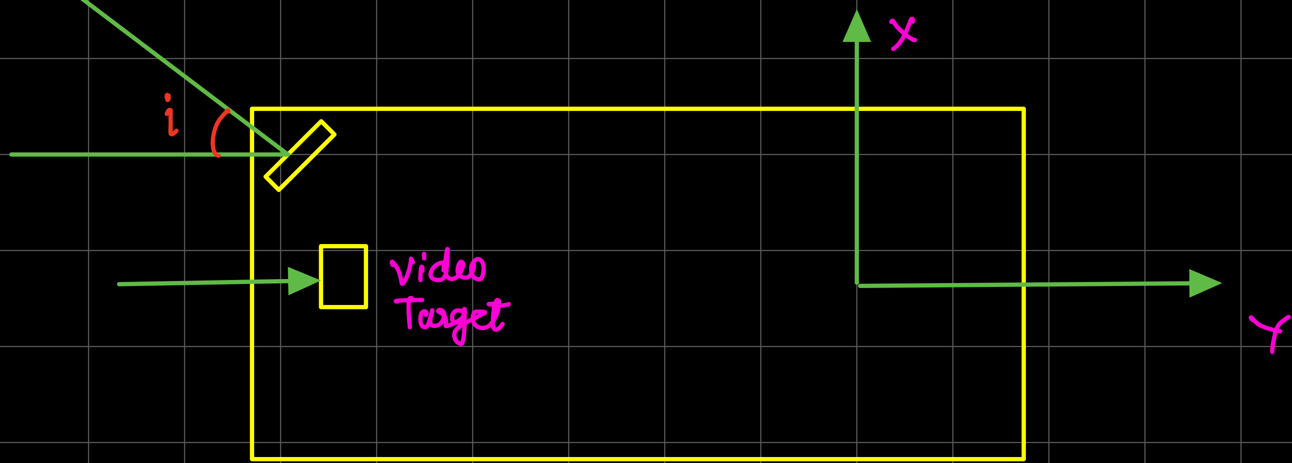
→ Peg System

→ Automated Guidance System



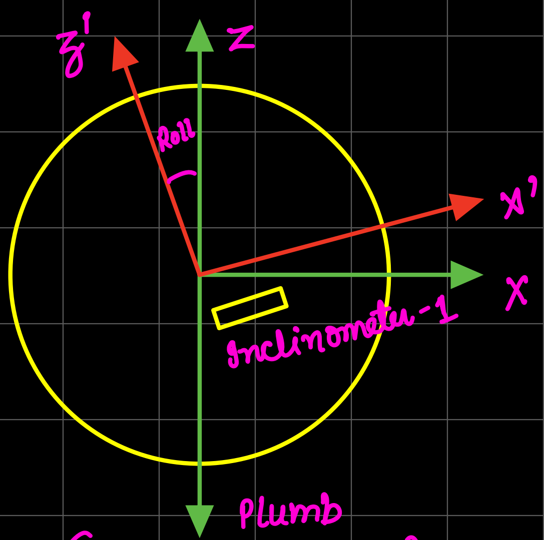
Peg System



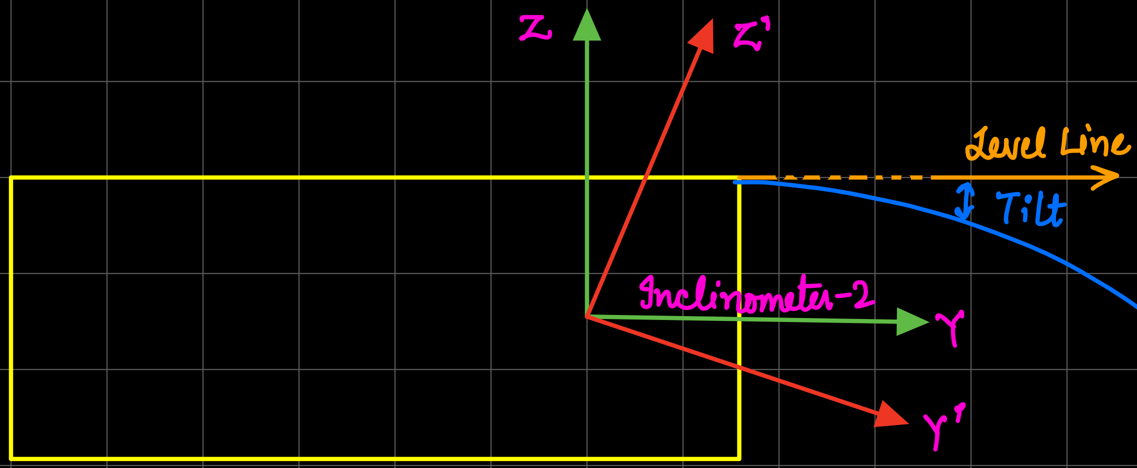


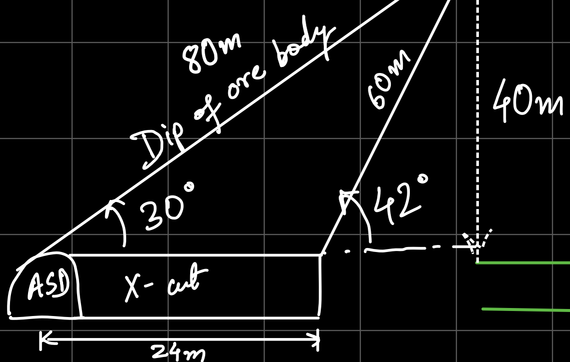
$i=0 \rightarrow$ Means moving in Y- direction

XY - Plane (Plan View)

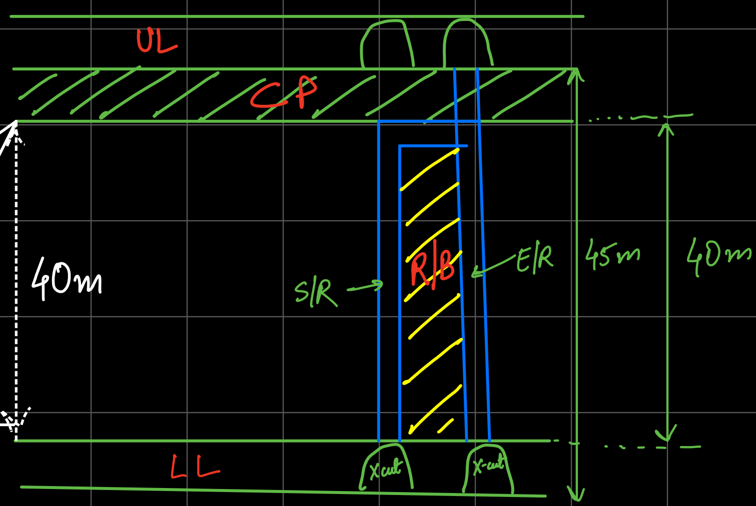


(Section View)

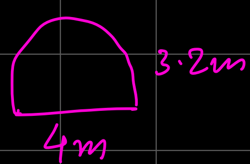




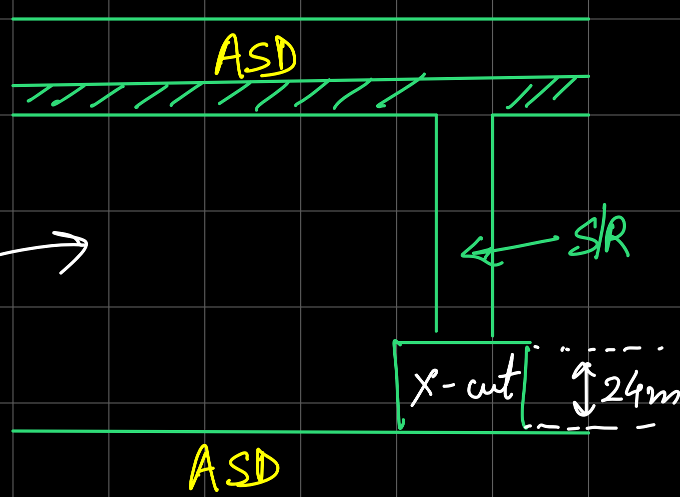
T- section



Front View



Plan View



Steps involved in its Development:

1) Preparatory Work
   (Compressed air pipeline, water pipe line, auxiliary fan + Vent. duct, Lighting, etc.)

2) Survey and Marking of S/R X-cut (if required)

3) Development of S/R X-cut in proposed direction as marked by surveyor.

4) Survey and Marking of S/R for its direction and Gradient (C/L and G/L)

5) Face marking + Face drilling + Charging & Blasting + Fume clearance + WS + LD + Face Preparation

6) Repetition of step 5 for next 2-3 rounds of blasting.

7) Partial mucking using LHD/LPDT.

8) Survey + Marking + Fixing pegs.

9) Platform preparation for next round of drilling (at least 2m behind the face) + Rope ladder fixing in the floor.

10) Face Marking + Face drilling + Support holes for platform + Rope ladder + Charging & Blasting.

11) (after dismantling the platform) + FC + WS + LD + face preparation + Platform preparation.

12) After 2-3 round of blasting, partial mucking will be done.

13) Development survey : after 15m development

14) Extending the rope ladder as face advances (Generally it is kept just behind the platform location)

15) Manhole preparation after 20-25 m progress of the face for keeping wooden planks + J/H M/C

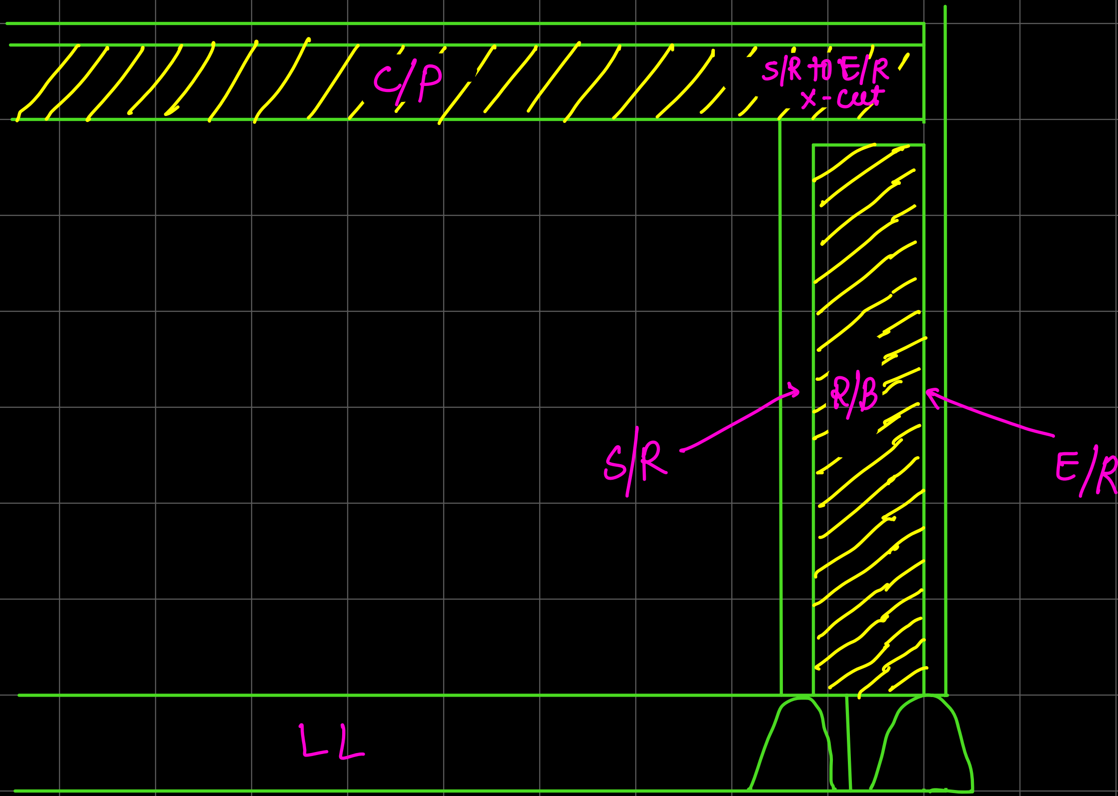
16) S/R to E/R x-cut Development (5m) in Horizontal plane.

Survey and Marking of C/L and G/L in proposed direction and gradient.

Next 2-3 round of blasting with the help of platform

Rest without platform

Blasted Material scrapping is an additional step in the x-cut development.



☆ Manpower distribution Plan ——

Sl No.	Activity	Shift (A)	Shift (B)	Shift (C)
1.)	Preparatory work	[$1_p + 2_h$ / $1_e + 2_h$] $+ 1_{PM} + 1_{MM}$	$1_e + 2_h + 1_{TM} + 1_{MM}$	
2.)	Survey & Marking of S/R X-cut	$1_s + 2_h + 1_{TM} + 1_{MM}$		Remark – (similar to drift development)
3.)	Development of S/R X-cut (24m)			
4.)	Survey & Marking of S/R for its direcⁿ & gradient	$1_s + 2_h$		
5.)	Face Marking + Face Drilling + C&B + FC	[$1_{JMO} + 2_h$ / $1_B + 1_h$] $+ 1_{TM} + 1_{MM}$		(For next 5 round of blasting)
6.)	WS + LD + Face preparation	$1_{TM} + 1_h + 1_{MM}$		(For next 5 round of blasting)

7.)	Survey + Marking + Fixing pegs (C/L & G/L)	1s + 2h		
8.)	Preparation of platform		1TM + 1h + 1MM	
9.)	Face marking + Face drilling + (C&B) + Dismantling of platform + FC	1 JMO + 2h , 1 TM 1B + 1h , 1MM		
10.)	WS + LD + FP + PP		1TM + 1h + 1MM	

★ **Main Activity (4' drilling) → 60m**

- ↳ Face marking + Face drilling + C&B + FC → 1 shift
- ↳ WS + LD + FP + PP → 1 shift

60 days

- ↳ S/R x-cut (24m) — 12 days
- ↳ S/R to E/R x-cut — 7 days

★ **Other Activities**

Manhole preparation — 3 days

Preparatory work — 2 days

Survey work — 8 shift / 3 days

87 days

Total time in days $= 87 + (87 \times 5\%)$

$= 92$ days

★ **Equipments / Machinery deployed :—**

- ↳ Jack Hammer
- ↳ Drill Jumbo

↳ LHD / LPDT

↳ Stage / Service Vehicle

↳ GEB

↳ Pipelines (Air + water)

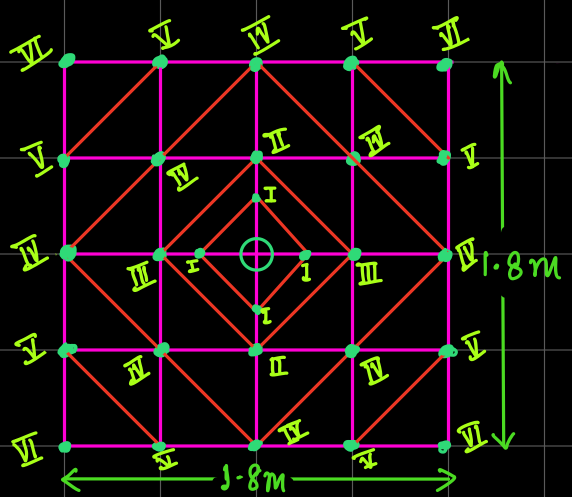
↳ Fan

↳ Ventilation duct

↳ Lightning

↳ Power Cable

☆ Drilling & Blasting —



→ Drill rod length = 2' & 4'

→ Drill bit dia. = 28 mm

→ Reamer bit dia. = 37 - 45 mm

→ Explosive / Emulsion type in the form of catridge

L = 10 cm

wt. = 100 gm

dia. = 20~25 mm

Total no. of holes = 29 + 1R

Total no. of blast holes = 28

Total Meterage (1.2 m depth) = $29 \times 1.2 + 1.2 \times 1$

= 36 m

Total explosive per round of blasting = 28×0.7 = 20 kg

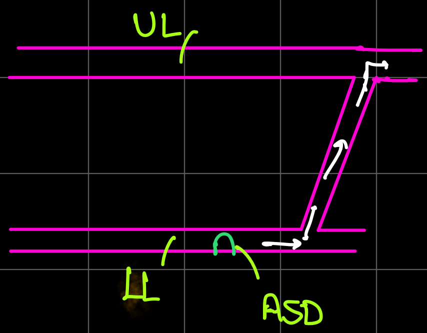
Total tonnage produced per round of blasting

$= 2 \times 2 \times 1.2 \times 0.9 \times 3$

$= 13$ tonnes

X Raise Ventilation

- Through pilot hole and forcing fan | Compressed air + forcing fan



Raise length is around (60m - 70m)

common practice in mine

Q. An end raise | ventilation raise of 70m length of opening size 2m × 2m is to be developed within two levels using Alimak Raise Climber. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plans, sections, figures) wherever needed.

1. Steps involved in raising
2. Manpower distribution plan manwise | shiftwise

③ Days required to complete raise

④ Equipments, Machinaries

⑤ Drilling and Blasting pattern

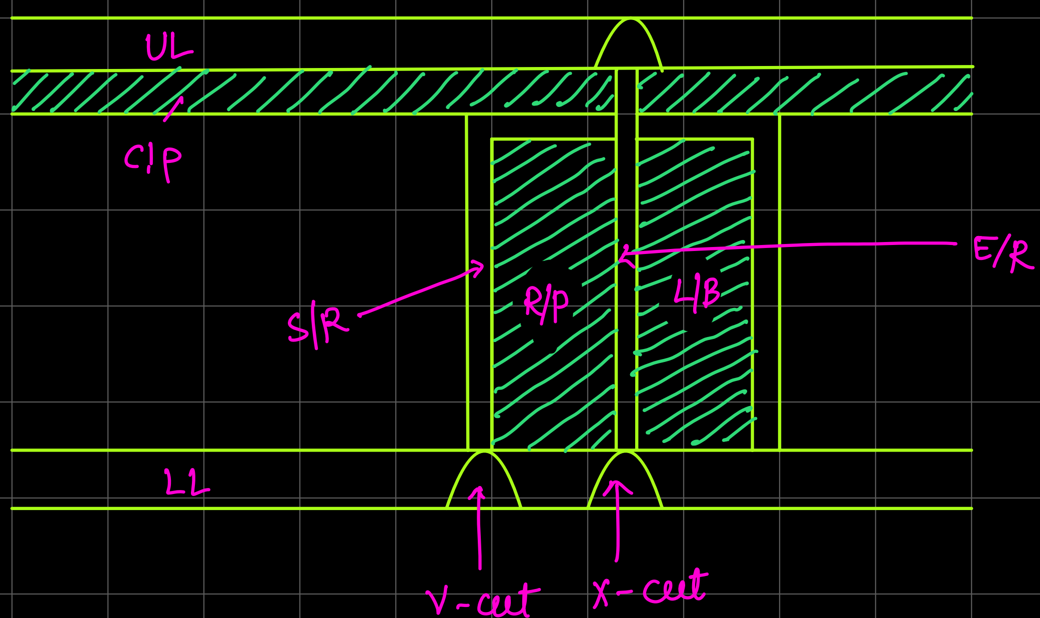
⑥ Drilling metrage per round of blasting

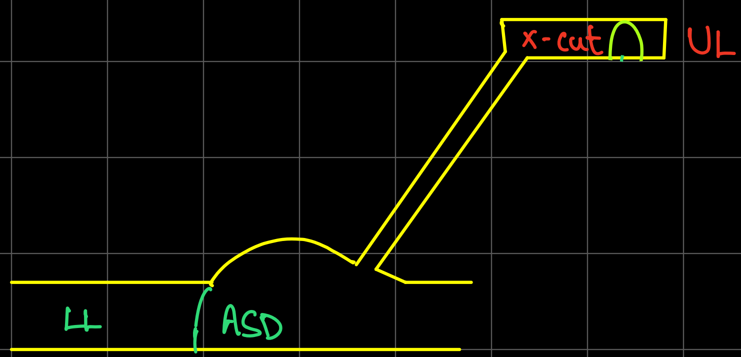
⑦ Explosive consumption " " "

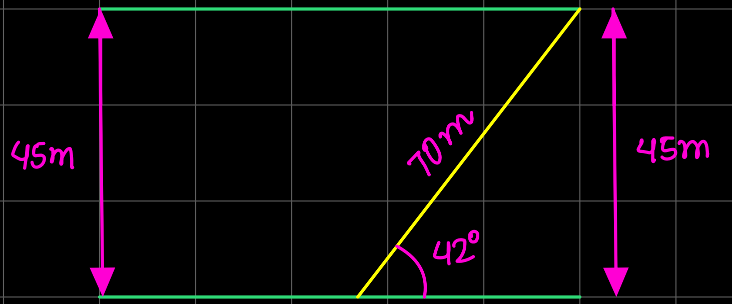
⑧ Tonnage produced per round of blasting

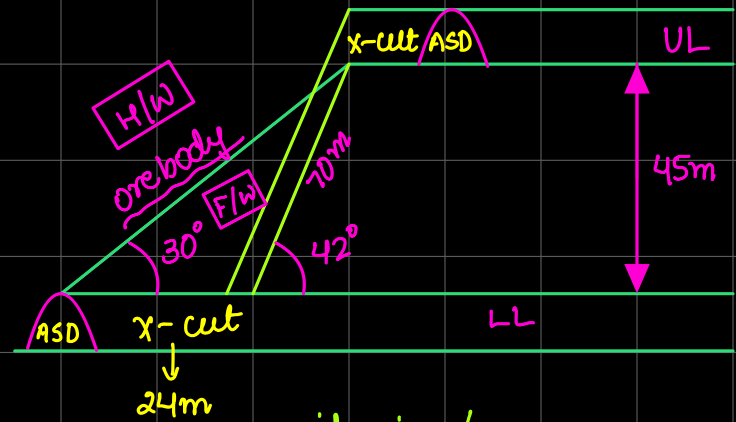
⑨ Powder factor obtained

⑩ Ventilation circuit for ventilating the face

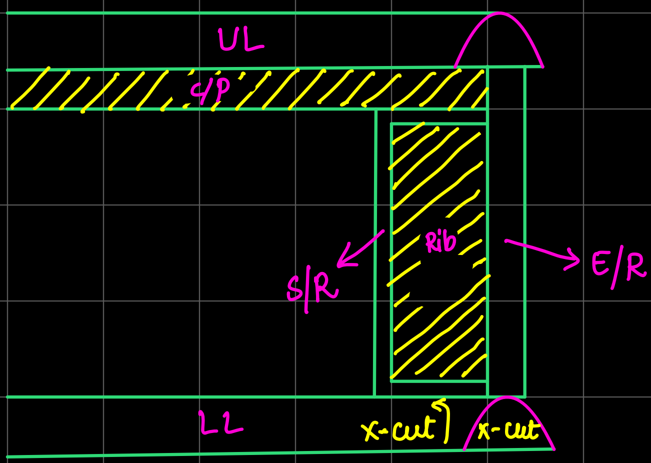




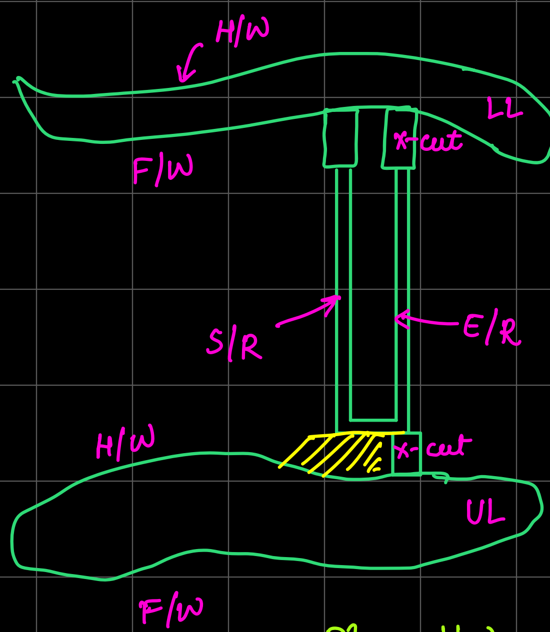




side view/ section view



Front view



Plan View

1. Steps Involved —

(i) Preparation work : water + compressed air pipe lines + LEB + lighting + Auxiliary fan + illumination + ventilation duct

(ii) Survey and marking (CL + GL) for x-cut

(iii) Development of x-cut upto the desired-length.

(iv) Heightening of back of x-cut upto 4~4.5m to accomodate ARC to facilitate mucking operation.

(v) Survey and marking for raise

(vi) 2~3 rounds of blasting in proposed direction and gradient

(vii) Shifting and Installation of ARC

→ Fixing angle guide

→ Service guide fixing

→ Cage Fixing

→ Composed air operated motor fixing

→ Hose Reel

→ Connection of hose reel with motor

→ Platform fixing

→ Canopy fixing

(viii) Face marking + face drilling + C & B + FC + WS & LD + FP + Repeatation above for 2~3 rounds of blasting.

(ix) Service guide rail extension of 2m + Face drilling (Incomplete)

(x) Face drilling + charging & Blasting + Fume clearance + WS + LD + Face preparation

(xi) Repeatition of cycle including service guide rail extension.

(xii) Partial mucking after 3-4 rounds of blasting.

(xiii) Development survey (CL & GL extension) and measuring opening size after 15~20m face advance.

(xiv) Uninstallation of Alimak Raise Climber (ARC).

(xv) Fitting of wooden ladder in the floor to make it operational.

For raising → per day 2 shifts only

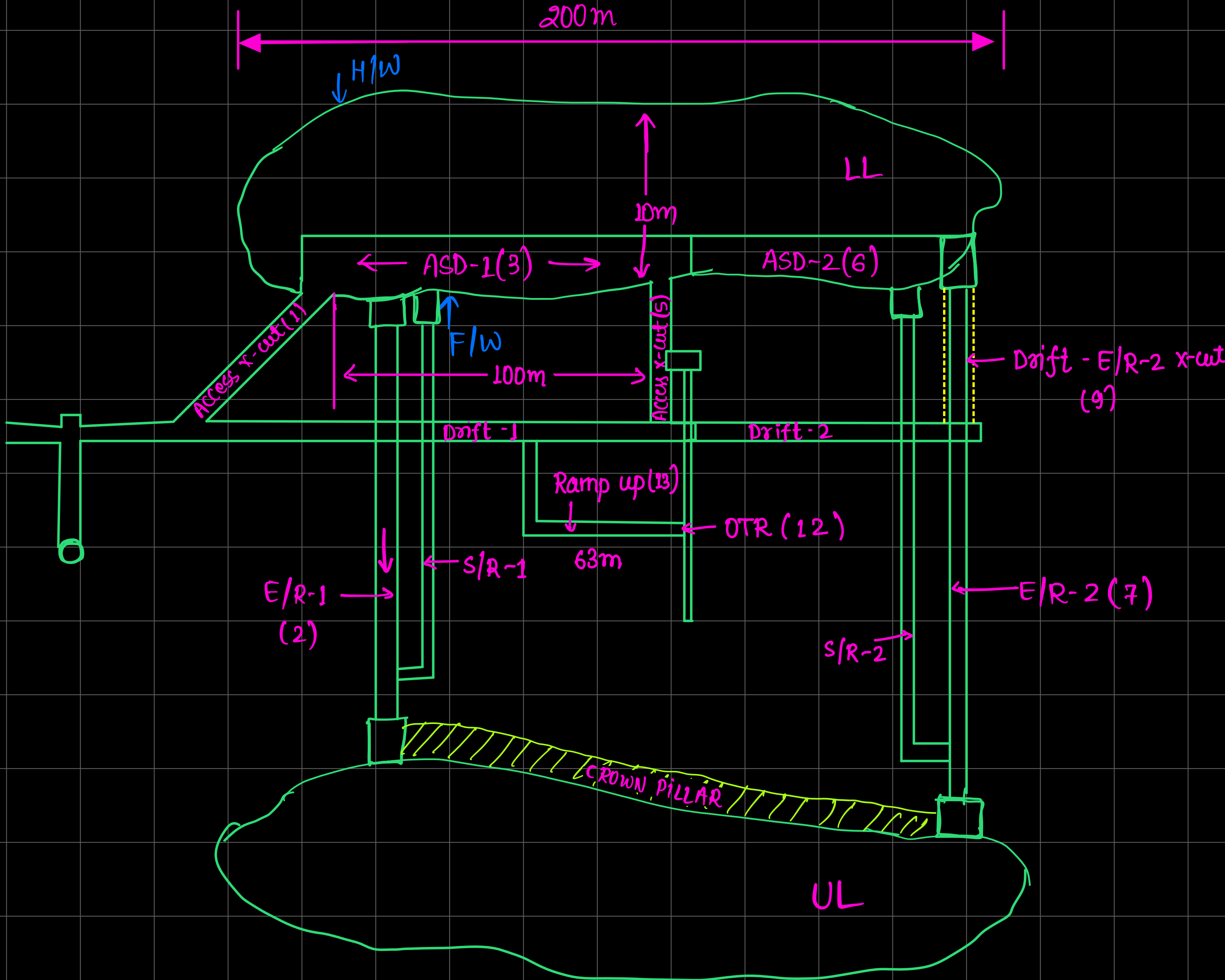
Total days required = 81 days

④ Machinery deployed

↳ same as conventional

→ Alimak Raise climber

⑤



1. Access x-cut (30~35m) → 18 days
2. E/R development (70m) → 81 days
3. Development of ASD (100m) → 53 days
4. Development of Drift (100m) → 53 days
5. Access x-cut (30-35m) → 18 days
6. ASD-2 (100m) → 53 days
7. E/R-2 Development (70m) → 81 days
8. Development of drift-2 (100m) → 53 days
9. Drift - E/R-2 x-cut → 12 days
10. S/R-1 development (60m) → 90 days
11. S/R-2 development (60m) → 90 days
12. OTR development (13m) → 13 days
13. Ramp up (63m) → 22 days

Total no. of days = 637 days